

Jonathan Bourget

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Education

BENG Bachelor of Electrical Engineering, Co-op Option — Carleton University

Sept 2024 – Present

- 3rd Year Standing, Contact for GPA | Carleton Academic Scholarship Recipient
- **Relevant Coursework:** Electronics I & II*, Switching Circuits, Physical Electronics*, Electromagnetic Fields*, Circuits & Signals (*in progress)
- **Expected Graduation:** April 2029

Hardware & Systems Projects

Avionics & Power Distribution PCB — C-130-Inspired RC Aircraft

Electronics Lead, Ongoing

- Lead power distribution and flight-controller electronics for a student RC aircraft team; built the project BOM and a CPM/Gantt schedule.
- Designed a multi-layer power distribution board in Altium Designer, splitting a 4S battery into raw motor power and regulated 12V, 5V, and 3.3V rails to isolate motor, logic, and sensor loads.
- Revised the design after peer review, specifying V2 changes: a buck converter in place of a linear 12V regulator, input bulk capacitance and spike suppression, per-regulator decoupling, and copper-pour high-current paths.
- Brought up a 2.4 GHz ExpressLRS radio link and a library-based CRSF receiver interface on STM32, verifying all 16 control channels.
- Bench-testing a BNO085 9-axis IMU on the STM32 flight controller ahead of flight-control firmware development.

Full-Stack AI Secretary — System Interfacing

Personal Project, Ongoing

- Architected a self-hosted assistant on an Orange Pi Zero 3 (ARM Linux) with a Node.js/Express and SQLite backend merging Gmail, Google Calendar, course deadlines, and live Yahoo Finance data into a prioritized daily brief.
- Deployed a React dashboard with a conversational interface over HTTPS via Tailscale; implemented with AI coding tools (Claude Code, Codex) from my own architecture and specifications.
- Built an ESP32-S3-driven HUB75 LED wall (192×32 px) that polls REST endpoints on the Pi over WiFi at two rates: a slow cycle for cached summaries and a fast cycle for live control.

Bicycle Network Infrastructure Proposal — RFID Feasibility Study

Team Project, 2026

- Conducted systems analysis for RFID-based smart bicycle infrastructure at Carleton University, researching RF technologies and evaluating bandwidth requirements against existing systems.
- Assessed user needs against feasibility constraints and synthesized findings into a formal proposal and executive presentation.

Electric Go-Kart — Full Cycle Design & Optimization

Two-Person Project, 2019

- Designed and built a 1.6kW electric drivetrain achieving 49 km/h top speed and 4+ hour runtime.
- Hand-welded the steel frame, integrated a 48V (4×12V) lead-acid battery pack, and replaced the throttle pedal to fix jerky acceleration.

Modular Bicycle Lighting System — Custom PCB Design

Personal Project, 2022

- Designed a weather-resistant RGBW lighting system with a custom Arduino-controlled PCB and 3D-printed enclosures, refined over several breadboard prototypes.

Work Experience

Lifeguard, Swim Instructor & First Aid Instructor — City of Ottawa

2021 - Present

- Held a safety-critical role requiring rigorous documentation, protocol compliance, and rapid decision-making in high-liability settings.
- Assessed dynamic situations, determined response procedures, and coordinated team actions under time pressure.
- Certified: Standard First Aid, CPR-C with AED, Airway Management & Oxygen Administration, SFA Instructor/Examiner

Technical Proficiencies

PCB & Circuit Design: Altium Designer, KiCad, schematic capture, multi-layer layout, power delivery, component selection

Embedded Systems & Microcontrollers: Arduino, STM32, ESP32, Orange Pi (Linux/ARM), real-time firmware

Programming: C/C++, Python, Verilog HDL, Java

Wireless & Communications: ExpressLRS, CRSF, RFID, wireless systems analysis, signal integrity considerations

Design & Simulation: Autodesk Fusion CAD, FPGA design (Vivado), MATLAB/Simulink, ANSYS Electronics (Fall 2026 coursework)

Lab & Fabrication: Soldering, oscilloscope, multimeter, bench supply testing, power supply specification, breadboard prototyping, hardware validation, welding, machining, 3D printing

Other: Git version control, LaTeX, technical writing, process documentation, requirements analysis

Languages: English, French (bilingual)